English

(1001CJA102922056)

Test Pattern

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
PART TEST
20-11-2022

JEE(Main+Advanced) : ENTHUSIAST COURSE [PHASE : TEAS, TOAS, TAAS, TNAS, I(A) & LIVE-I(A)] (SCORE-I)

Time : 3 Hours **Maximum Marks : 198**

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble ◯ **COMPLETELY.**
12. The correct way of darkening a bubble is as : ●
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **g = 10 m/s²** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS	
Atomic No. :	H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58
Atomic masses :	H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca=40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba=137, Ce = 140

• **Boltzmann constant**	$k = 1.38 \times 10^{-23}\ JK^{-1}$
• **Coulomb's law constant**	$\frac{1}{4\pi\varepsilon_0} = 9\times10^9$
• **Universal gravitational constant**	$G = 6.67259 \times 10^{-11}\ N\text{–}m^2\ kg^{-2}$
• **Speed of light in vacuum**	$c = 3 \times 10^8\ ms^{-1}$
• **Stefan–Boltzmann constant**	$\sigma = 5.67 \times 10^{-8}\ Wm^{-2}\text{–}K^{-4}$
• **Wien's displacement law constant**	$b = 2.89 \times 10^{-3}\ m\text{–}K$
• **Permeability of vacuum**	$\mu_0 = 4\pi \times 10^{-7}\ NA^{-2}$
• **Permittivity of vacuum**	$\in_0 = \frac{1}{\mu_0 c^2}$
• **Planck constant**	$h = 6.63 \times 10^{-34}\ J\text{–}s$

Space for Rough Work

PART-1 : PHYSICS

SECTION-I : (Maximum Marks: 24)

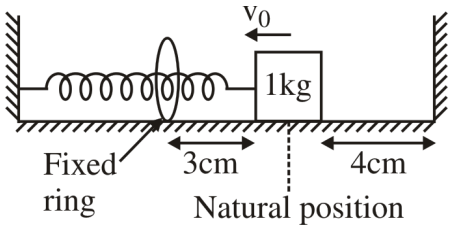
- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : –2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

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1. Spring is in its natural length. Collision of the block with the ring fixed with the horizontal surface and the wall is perfectly elastic. If at t = 0, a velocity v_0 = 0.5 m/s is given to the block and spring constant of spring is 100 N/m, mass of block is 1 kg, then :-



(A) 0.3 m/s is velocity of block just after collision with ring.

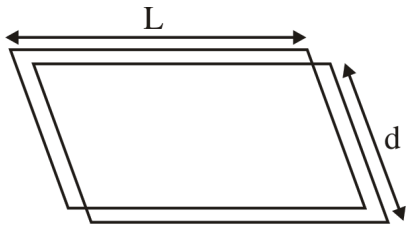
(B) 0.4 m/s is velocity of block just after collision with ring.

(C) $\pi/10$ sec is time period of the motion.

(D) 0.4 m/s is velocity of block just before collision with wall.

Space for Rough Work

2. A long metal strip of thickness t = 1 mm and width d = 1 cm carries current along its length (L). The current in the metal strip is following towards the left. It is placed in a uniform transverse magnetic field B = 2T (as shown in figure). The potential difference recorded across the width 'd' of the plate is found to be 3×10^{-6} volt. Select the correct statement.



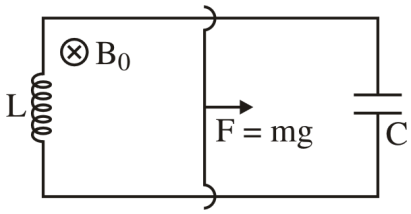
(A) Electric field resulted across the width 'd' is 1×10^{-3} V/m.

(B) Electric field resulted across the width 'd' is 3×10^{-4} V/m.

(C) Magnitude of electric force on the electron due to the electric field that is resulted due to the motion of the electron inside metal strip is 4.8×10^{-22} N.

(D) The drift velocity of the electron is 1.5×10^{-4} m/s.

3. A conducting rod of mass 'm' is placed on frictionless horizontal rails and a force of "mg" starts acting on the conducting rod at t = 0 toward right perpendicular to rod. One end of the parallel conducting rod is connected by the inductor of inductance 'L' and other end of the conducting rod is connected by the capacitor as shown in the diagram. A uniform vertical magnetic field exist in the region perpendicular to the plane of the ring. (take $CB^2\ell^2$ = mass of rod)



(A) The maximum horizontal displacement of the rod is $\frac{4mgL}{B^2\ell^2}$.

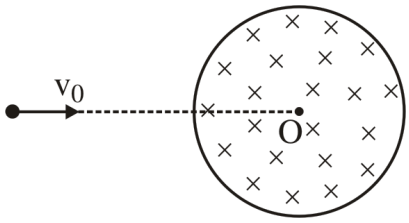
(B) The maximum horizontal displacement of the rod is $\frac{MgL}{B^2\ell^2}$.

(C) The angular frequency of the oscillation of the rod is $\frac{B\ell}{\sqrt{2}mL}$.

(D) None

Space for Rough Work

4. A circular region of radius R = $\sqrt{3}$m which has a uniform magnetic B = 0.2 Tesla directed into the plane of the figure. A particle having mass m = 2gm, speed v = 0.3 m/s and charge q = 1mC is projected along the radius of the circular region as shown in diagram. Select the correct statement(s).



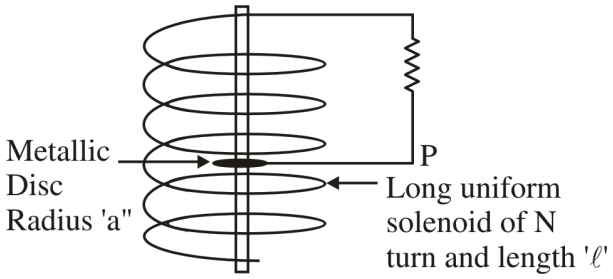
(A) Deviation produced in the charge particle is $\frac{\pi}{3}$.

(B) Deviation produced in the charge particle is $\frac{\pi}{2}$.

(C) Average acceleration of the charge particle is 9π cm/s^2.

(D) Distance travelled by the particle in the magnetic field is π m.

5. A metallic disc of radius 'a' is mounted on a rigid axle, is rotating with constant angular velocity ω inside a long solenoid of inductance L whose two ends are connected to the rotating disc of by two brush contact as shown. Resistivity of the wire of solenoid is ρ and resistance R is connected between solenoid and metallic disc as shown in figure. Then :-



(A) The induced e.m.f. between terminals P and Q is $\frac{\mu_0 N i \omega a^2}{2\ell}$ when current in the coil is i.

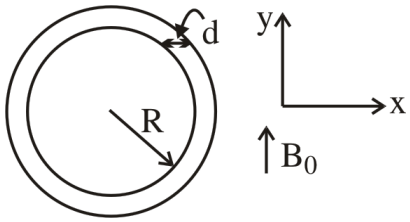
(B) Current in the coil increases or decrease with time depending upon the value of ω.

(C) ω_{min} for increasing the current in coil is $\frac{\ell R}{2\mu_0 N \rho a^2}$.

(D) ω_{min} for increasing the current in the coil is $\frac{2\ell R}{\mu_0 N a^2}$.

Space for Rough Work

6. A thin walled non-magnetic spherical shell with wall thickness d and radius R (R >> d) is placed in a uniform magnetic field $\vec{B} = B_0(\hat{j})$, which slowly changes as $\frac{dB}{dt} = \alpha$, where α is a positive constant. Electrical resistivity of the material of the shell is 'ρ'. Then select the correct option.



(A) Magnetic moment induced in the shell is in negative 'y' direction.

(B) Magnetic moment induced is directly proportional to d.

(C) Magnetic moment induced is inversely proportional to d.

(D) Magnetic moment is independent of the d.

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

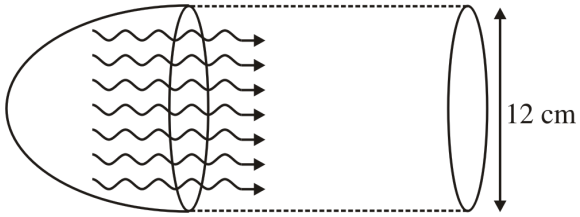
- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

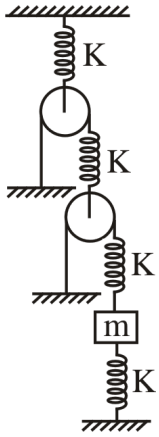
Zero Marks : 0 In all other cases.

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1. A pulse source of microwave produces burst of 20 GHz radiation with each burst having a duration 1 ns. A parabolic reflector (R = 6 cm) is used to produce a parallel beam of radiation as indicated in the diagram. The average output power of the source during each pulse is 25 kW. If microwave travel in air, maximum value of electric field E is $x \times 10^4$ V/m. Then find the value of x.

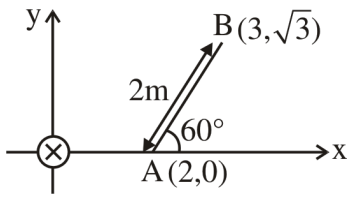


2. Four identical massless springs of the spring constant 70 N/m, two massless pulley and a block of mass of 6.6 kg are connected as shown and system is kept in vertical plane. Initially the system is in equilibrium. Now the block is slightly displaced from equilibrium position of equilibrium position along a vertical line and then released. Find the time period (in sec).



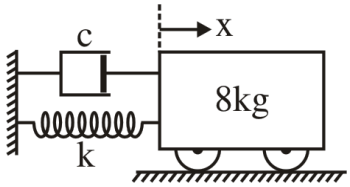
Space for Rough Work

3. An infinity long current carrying wire is placed at the origin along the z-axis. Current in the wire is 2 amp. A straight wire AB of length 2m is placed from A(2,0) to B$(3, \sqrt{3})$ as shown in diagram. Then the value of $\int_A^B B \cdot d\ell$ is $x \times 10^{-8}$ (in S.I. Unit). Find the value of x.

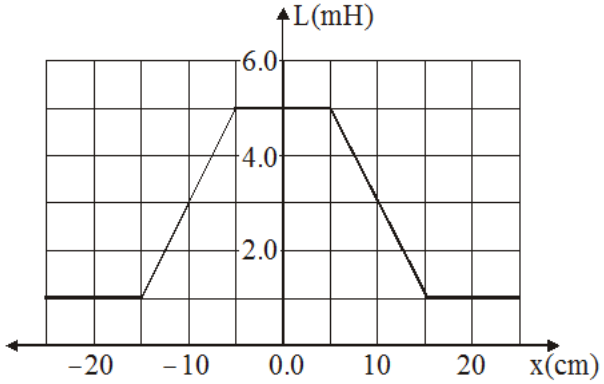


4. Suppose Bundi is a small town near Kota with a demand of 1200 kW electric power at 220 Volt is located 30 km from KTPS (Kota Thermal Power Station) generating power at 440 Volt. The resistance of the two vires line carrying power is 0.2 Ω/km. Bundi gets power from the line through a 4000 V – 220 Volt step down transformer at substation in Bundi. Assuming no power loss in the transformer. Power supplied by the KTPS to Bundi substation is :- (in kW)

5. The 8-kg body is moved 0.2 m to the right of the equilibrium position and released from rest at time t = 0. Determine angular frequency of the damped oscillation. (Given that) The viscous damping coefficient c is 20 N·s/m, and the spring stiffness k is 32 N/m.



6. A coil is wound on a hollow insulating cylinder, which contains in it a laminated iron core. The dependence of inductance L with the displacement x of the iron core is shown in the figure. In the initial state x = 0 (the core is fully inserted into the coil), the current in the coil is 1.0 A. Find current in the coil immediately after the core is quickly taken out of the coil. (Answer in Amperes)



Space for Rough Work

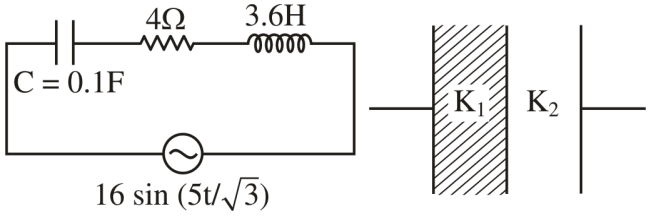
SECTION-III : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

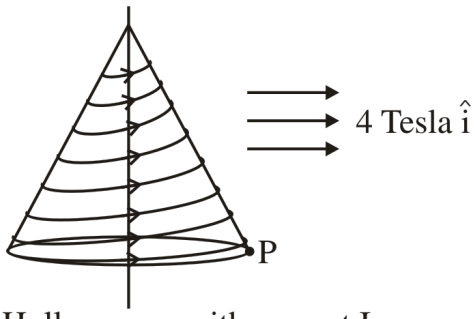
Full Marks : +3 If only the correct answer is given.
Zero Marks : 0 If no answer is given.
Negative Marks : –1 In all other cases

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1. The circuit shown in figure is driven by an AC voltage 16 sin $\frac{5t}{\sqrt{3}}$ volts. The capacitor (area of each plate of capacitor = 10^{-4} m^2) is suddenly removed at t = $\frac{7\pi}{10\sqrt{3}}$ seconds and then the geometrically, identical dielectric slabs of dielectric constant of $K_1 = 2$ and $K_2 = 6$ respectively. Find surface charge density at the interface in kC/m^2. If your answer is α then find the value of α/150.



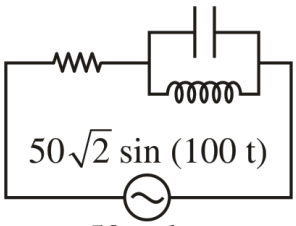
2. A hollow conducting cone has mass 10 πgm and height 20 cm and radius 5 cm. A current I flowing uniformly along its curved surface area. This cone is kept on a very rough surface and a uniform magnetic field B = 4 tesla $(\hat{i})$ exist in space. The maximum value of I (in Amp) so that the cone does not topple about point P is :-



Hollow cone with current I_0

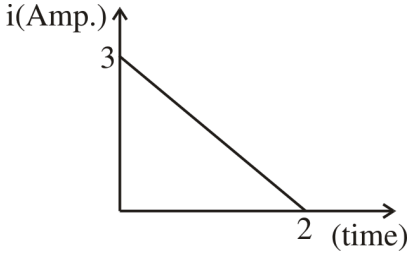
Space for Rough Work

3. In the AC circuit shown in the diagram. L = 80 mH and C = 2.5 mF and R = 6Ω. The rms voltage of AC source is 50 volt of frequency 100 rad/sec. The rms current (in amp) through the capacitor is I_0. Find the value of $I_0/2$.

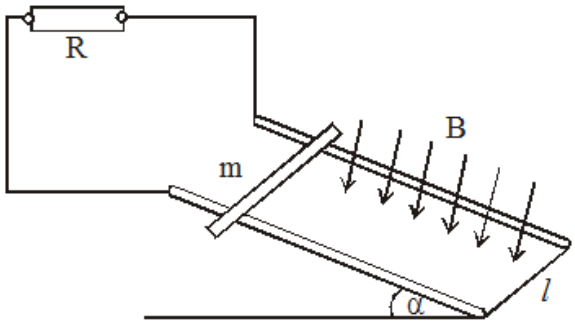


50 volt,
100 rad/sec

4. When magnetic flux through a coil is changed the variation of induced current in the coil with time is as shown in the graph. If resistance of coil is 2Ω, then the magnitude of total change in flux (in weber) of coil will be :

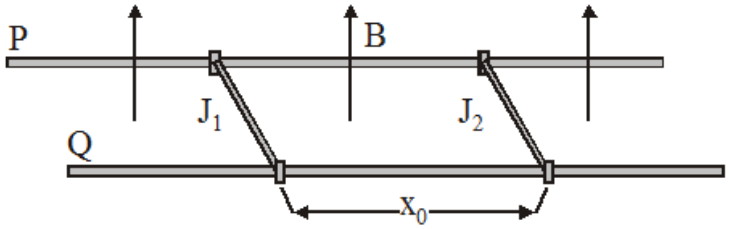


5. A homogeneous field of magnetic induction B is perpendicular to a track of gauge l which is inclined at an angle α = 30° to the horizontal. A frictionless conducting rod of mass m straddles the two rails of the track as shown in the figure. If maximum speed of the conductor is = $\frac{mgR}{nB^2l^2}$. Then find n



Space for Rough Work

6. Two long conducting bars P and Q are fixed parallel to each other in a horizontal plane some distance apart. On these bars, two parallel jumpers J_1 and J_2 each of resistance R are arranged at separation x_0 as shown in the figure. The jumpers can slide on the bars without friction. At some point of time, a uniform vertical magnetic field of induction B is switched on. If the final separation between the jumpers is $\frac{x_0}{N}$, find N. The force between the jumpers due to their currents is negligible as compared to the force of interaction between current in the jumper and the magnetic field. (Self inductance of system is neglected)



Space for Rough Work

PART-2 : CHEMISTRY

SECTION-I : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
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Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.

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Negative Marks : –2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

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1. Which of the following statement(s) is/are CORRECT ?

(A) Aspirin is an example of non-narcotic analgesics.

(B) Soap is formed by boiling triglyceryl stearate with caustic soda

(C) Saccharin and alitame have S atom in their structure.

(D) In HDPE preparation, Zeigler natta catalyest is used up

2. Which of the following statement(s) is/are CORRECT ?

(A) Vitamin C is water soluble

(B) Cytosine, Thymine and Uracil are bases derived from purine.

(C) A nucleic acid is a polynucleotide by condensing sugar, phosphate and nitrogeneous base where the polymeric link is a phosphodiester.

(D) Adenine and thymine are paired by two hydrogen bonds whereas Guanine and Cytosine are paired by three hydrogen bonds.

Space for Rough Work

3. Which of the following statement(s) is/are NOT correct ?

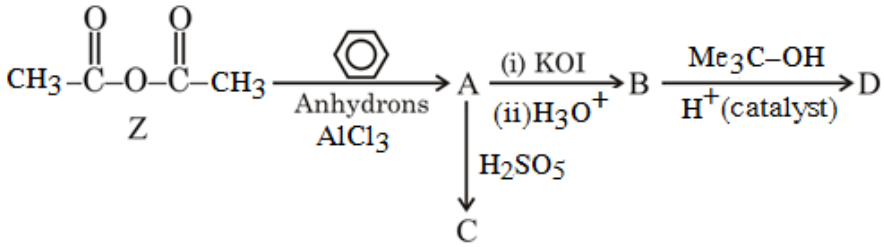
(A) Duma's method is not applicable to compounds containing nitrogen in nitro and azo groups.

(B) Sulphur is determined in an organic compound by heating it in a carius tube with Na_2O_2 and precipitating all sulphur as $BaSO_4$.

(C) For determining phosphorus, the organic compound is heated with conc. HNO3 which oxidizes all phosphorus to H_3PO_4. It is precipitated as ammonium phosphomolybdate by adding NH_3 and ammonium molybdate.

(D) Oxygen is estimated by decomposing the organic compound by heating it in a stream of N_2 gas. The gaseous mixture is passed over red hot coke which converts all O_2 to CO. The CO produced is treated with warm I_2O_5 which produces CO_2 and I_2. The CO_2 produced gives estimation of oxygen.

4. Consider the following sequence of reactions :



A gives positive DNP test. B gives positive $NaHCO_3$ test.
Which of the following is/are matched correctly ?

(A) A on reaction with NaOD in D_2O shows an increase of 3 amu in molecular mass.

(B) The solution of B in benzene gives colligative properties with van't Hoff factor greater than 1.

(C) Z on reaction with benzaldehyde in presence of CH_3COONa followed by acidification gives a product with molecular mass 148 amu.

(D) The pattern of hydrolysis of C and D in NaOH solution can be different

Space for Rough Work

5. Which of the following reaction(s) will result in formation of $Me_3C–NH–CH_3$

(A) $Me_3C–NH–\overset{\overset{O}{\|}}{C}–H \xrightarrow[(ii)\ H_2O]{(i)\ LAH,\ dry\ ether}$

(B) $Me_3C-NH_2 \xrightarrow{CHCl_3+KOH} \xrightarrow[(ii)\ H_2O]{(i)\ LAH,\ dry\ ether}$

(C) $Me_3C–\underset{\underset{O}{\|}}{C}–CH_3 + CH_3NH_2 \longrightarrow$

(D) $Me_2C=CH_2 \xrightarrow{CH_3NH_2}$

6. Consider the following compounds :

(I) CCl_3CHO (II) $CH_3–\overset{\overset{O}{\|}}{C}–H$ (III) $CH_3–\overset{\overset{O}{\|}}{C}–CH_3$ (IV) $Ph–\overset{\overset{O}{\|}}{C}–CH_3$

Select the CORRECT statement(s) :

(A) Extent of hydration : I > II > III > IV

(B) II & III on reaction with $CaOCl_2$ solution followed by heating of the Ca containing moeity give back the respective starting compound.

(C) All compounds show positive iodoform test.

(D) III is more polar than II.

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

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- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

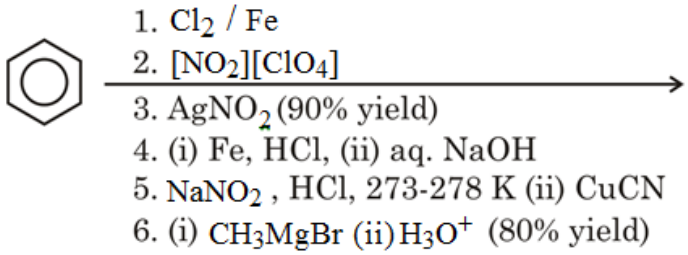
Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

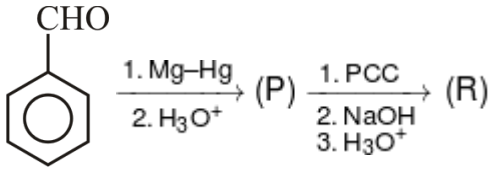
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1. In the following roadmap, 78 g of benzene is used at the start. Percentage yields, wherever applicable, are indicated in brackets. At the end of all the steps, what mass of the product is obtained ?

(Atomic mass data : C = 12, H = 1, N = 14, O = 16, Cl = 35.5, Br = 80)



2.



Given the ratio of the molar mass of R in g/mol to the degree of unsaturation present in the compound

3. A 2L, 0.1 M solution of sodium acetate is electrolyzed using Pt electrodes. If a current of 9.65 mA is passed for 2500 seconds, and the current efficiency is 80%, then the mass of hydrocarbon in mg produced at the anode will be-

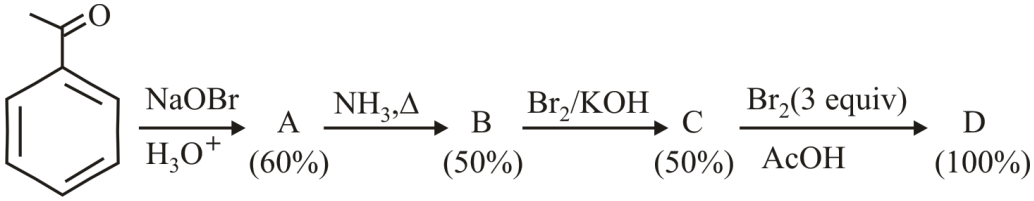
(1 faraday = 96500 C)

Space for Rough Work

4. The number of chiral center present in maltose

5. Among the twenty naturally occuring amino acids, if
a = number of amino acids whose pH at isoelectric point is < 4.
b = number of amino acids whose pH at isoelectric point is > 7.
c = number of amino acids which have an aromatic ring in their structure.
Find the value of expression $a \times \frac{b}{c}$.

6. In the following reaction sequence, the amount of D (in g) formed from 10 moles of acetophenone is____.
(Atomic weight in g mol–1: H = 1, C = 12, N = 14, O = 16, Br = 80. The yield (%) corresponding to the product in each step is given in the parenthesis)



Space for Rough Work

SECTION-III : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks	:	+3	If only the correct answer is given.
Zero Marks	:	0	If no answer is given.
Negative Marks	:	–1	In all other cases

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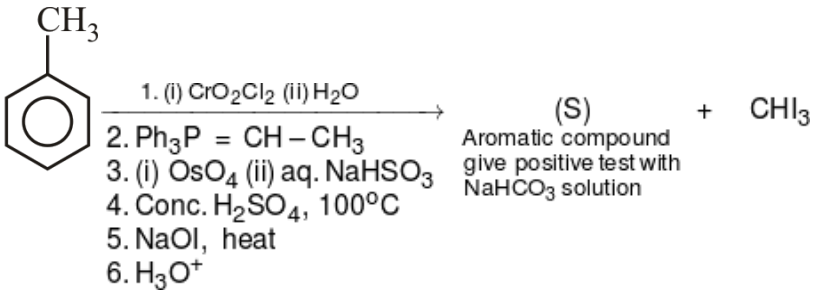
1. Consider the heating reactions of dicarboxylic acids, $HOOC–(CH_2)_x–COOH$ where x is the number of methylene groups between the COOH groups varies from 1 to 5.

   p = number of acids in which one of the products is CO_2

   q = number of acids in which one of the products in H_2O

   r = number of acids in which products contain both CO_2 and H_2O

   Calculate (p + q – r)

2. 

   If the molar mass of S is x g/mol, report x/17.

3. How many of the following drugs are bactericidal in nature?

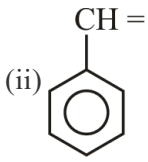
   Erythromycin, tetracycline, penicillin, ofloxacin, chloramphenicol.

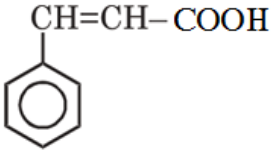
Space for Rough Work

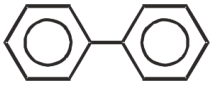
4. How many of the following are addition polymers ?
Polyethene, Teflon, PVC, Nylon, Bakelite, Melamine-formaldehyde resin, PMMA, Polystyrene, Natural rubber, Urea-formaldehyde resin.

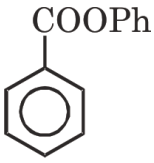
5. In how many of the following compounds is the reactivity greater than that of benzene towards nitration via electrophilic aromatic substictution mechanism.

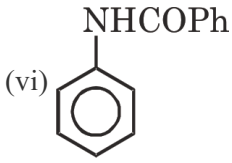
(i) 

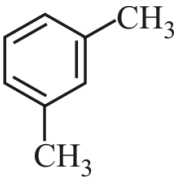
(ii) 

(iii) 

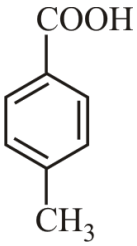
(iv) 

(v) 

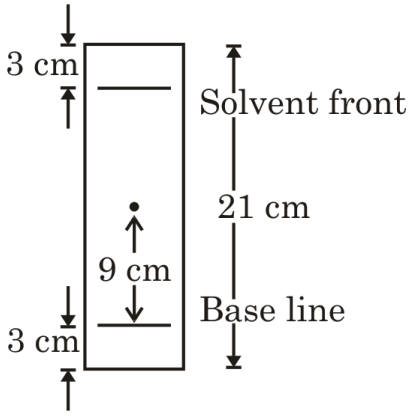
(vi) 

(vii) 

(viii) 

(ix) 

6. In a developed chromatogram of a flower extract shown, find the retardation factor ***(x)*** of the segment. Report 5***x***.



Space for Rough Work

PART-3 : MATHEMATICS

SECTION-I : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : –2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

---

1. The coffiecient of x^6 in the expansion of $\left(1+x^2\right)^3\left(1-\sqrt[3]{x}\right)^7\left(1+\sqrt[3]{x}\right)^4$ is :

(A) 49 (B) –43 (C) 43 (D) 0

2. 15 boys and 5 girls are to be seated in a row randomly. The probability that between any two girls atleast 2 boys sit is

(A) $\frac{6!\ 7!}{20!\ 19!}$ (B) $\frac{12!\ 15!}{6!\ 20!}$ (C) $\frac{15!\ 12!}{7!\ 20!}$ (D) $\frac{16!\ 18!}{20!\ 7!}$

Space for Rough Work

3. The value of $\frac{1}{2^{15}}\sum_{r=0}^{15}(-1)^r 5^{15-r}\ {}^{30}C_r\ {}^{30-r}C_{15-r}$ is

(A) number of ways of selection of 15 persons from 30 married couples such that no husband is selected with his wife

(B) number of ways of selection of 15 persons from 15 married couples

(C) number of ways of writing answers to 15 true-false type questions selected out of 30 questions

(D) number of ways of writing answers to 15 multiple choice correct (one or more choice correct) questions selected out of 30 questions. (each question has 4 choices)

4. Consider a set 'S' of 15-digit numbers ($d_i \in \{0, 1, 2,.....9\}$)
$S = \{d_1d_2d_3.....d_{14}d_{15} \mid d_i \le d_{i+1}\}$ for all $i \in \{1, 2,.....14\}$
Let n(S) denotes the number of elements in S, then

(A) exponent of 3 in n(S) is 3

(B) exponent of 3 in n(S) is 1

(C) unit digit in $(n(S))^4$ is 0

(D) unit digit in $(n(S))^4$ is 6

5. Which of the following statement(s) is/are **INCORRECT** ?

(A) $\log_{(\sqrt{6}-\sqrt{5})}(\sqrt{8}-\sqrt{7}) > 1$

(B) $\log_{(\sqrt{6}-\sqrt{5})}(\sqrt{8}-\sqrt{7}) < 1$

(C) $\log_7\left(\sqrt[3]{2}+\sqrt[3]{5}\right)\left(\sqrt[3]{4}-\sqrt[3]{10}+\sqrt[3]{25}\right)$ is an integer value

(D) number of rational terms in the expansion of $\left(7^{\frac{1}{7}}+2^{\frac{1}{5}}\right)^{105}$ is 3

6. If 'x' satisfies the equation $\sum_{r=1}^{2023}\sin(rx)\cos(2024-r)x = \frac{2023}{2}$ then

(A) $\sin 6\theta = \sin(2024x)$ is true for 6 real values of θ in $[0, 2\pi]$

(B) $\sin 6\theta = \sin(2024x)$ is true for 12 real values of θ in $[0, 2\pi]$

(C) $\sin(2024x) > \cos(2024x)$

(D) $\sin(2024x) < \cos(2024x)$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

---

1. Let 'a' be the number of ways in which 3 consecutive persons can be selected from n persons sitting around a round table and 'b' be number of ways of selecting one male and one female from n + 3 males and n + 3 females, then the value of $36\left(\sum_{a=4}^{2023}\frac{a+3}{(a+1)!\ .\ b}+\frac{1}{2026!}\right)$ is

2. Number of value(s) of x in $[0, 2\pi]$ satisfying the equation

$$\left(\log_{(\sqrt{2}-1)}\tan x\right)\sqrt{\log_{(\sqrt{2}-1)}\sqrt{(3-2\sqrt{2})(\sqrt{2}-1)}+\log_{\tan x}(\sqrt{2}-1)}=-1 \text{ is}$$

3. Number of real solution(s) of the equation $\frac{\sin^2 2\theta}{2}+6\sin\theta.\cos\theta(\sin\theta+\cos\theta)=10$ in $[0, 4\pi]$ is

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Space for Rough Work

4. If $\left(\log_p q + \log_q p\right)^2 = 2029$, $p > q > 1$, then the number of values of $\theta \in (0, 6\pi)$ satisfying the equation $\frac{1}{\log_{pq} q} - \frac{1}{\log_{pq} p} = \text{cosec}\theta$ is

5. Two friends 'A' and 'B' plays a game by picking one ball alternately without replacement from a bag containing 7 white and 4 red balls in the order ABABAB..... The one who gets a first red ball wins the game. The probability that A got two attemps to draw balls given that B wins the game is $\frac{m}{n}$ (in lowest terms, $m, n \in I$) then $n - m$ is

6. A random variable X has the probability distribution

X	1	2	3	4	5	6	7	8	9
P(X)	0.11	0.2	0.14	0.26	0.13	0.01	0.03	0.04	0.08

for the events A = {X is an odd number},

B = {X is the exponent of 13 in 100!},

C = {X is unit digit of 7^n, $n \in N$}, the probability $P((A \cap B) \cup C)$ is

Space for Rough Work

SECTION-III : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks	:	+3	If only the correct answer is given.
Zero Marks	:	0	If no answer is given.
Negative Marks	:	–1	In all other cases

---

1. Let X be a binomial variate with mean and variance 30 and 5 respectively and Y be a binomial variate with mean and variance 9 and $\frac{27}{4}$ respectively. Let $P_1(X = r)$ and $P_2(Y = r)$ deontes the probabilities of obtaining 'r' success for random variables X and Y respectively. Let a, b be two positive integers (different from one) satisfying

$a^{a+b} = b^{12}$ and $b^{a+b} = a^3$ and if $\frac{P_1(X = a)}{P_2(Y = b)} = \frac{11.17.2^{35} \cdot p^4}{q^{70}}$ then p – q is

2. Consider the following sets

$A = \{\theta : \sec\theta + \text{cosec}\,\theta = \sin\theta + \cos\theta, \theta \in [0, 2\pi]\}$,

$B = \{\theta : \cos\theta = \sqrt{1 - \sin\theta}\ , \theta \in [0, 3\pi]\}$,

$C = \left\{\theta : 2^{\sin^2\theta} > \frac{\cos\theta}{2\cos\theta + 1}, \theta \in R\right\}$. The number of elements in the set $(A \cup B) \cap C$ is

3. Number of real solution(s) of equation $\frac{1}{3}\log_2(\cos^3 5x + \cos^3 7x) = 1 + \log_2\cos 6x + \log_2 \cos x$ in $[0, \frac{\pi}{2}]$ is

---

Space for Rough Work

4. Consider the observations

$x_i = \frac{5}{2} + p,\ i = 1 \text{ to } 20,$

$x_i = \frac{7}{2},\ i = 21 \text{ to } 40,$

$x_i = \frac{5}{2} - p,\ i = 41 \text{ to } 60,$

if the variance of these 60 observations is $\frac{4}{9}$ then the value of $6p^2$ is

5. Let $f(m) = \sum_{r=1}^{m} \frac{(m^r + m^{r-1})\left(1 - \frac{1}{r}\right)}{{}^{m}C_r}$ then the unit digit of the value of f(2023) is

6. Let N be the number of ways in which a kid can arrange 5 balls, one each of red, green, black, yellow and pink in a row such that, yellow is not adjacent to black and pink. Green is not adjacent to black and red. The value of $\frac{2}{7}$ N is

Space for Rough Work

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME
(Academic Session : 2022 - 2023)

JEE (Advanced)
PART TEST
20-11-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TEAS,TOAS, TAAS, TNAS, I(A) & LIVE-I(A)] (SCORE-I)

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	B,C	A,D	B,C	A,C,D	A,B,D	A,B
SECTION-II	Q.	1	2	3	4	5	6
	A.	4.08	1.88	2.09 to 2.10	2280.00	1.50 to 1.56	5
SECTION-III	Q.	1	2	3	4	5	6
	A.	1	1	5	6	2	2

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6
	A.	A,B,C,D	A,C,D	A	A,C,D	A,B	A,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	116.64	25.33	3.00	10.00	1.50	495.00
SECTION-III	Q.	1	2	3	4	5	6
	A.	5	8	2	6	4	3

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	C	C	A,C	B,D	B,D	A,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	0.05	2.00	0.00	6.00	19.00	0.36
SECTION-III	Q.	1	2	3	4	5	6
	A.	2	5	3	2	6	4

HINT – SHEET

PART-1 : PHYSICS

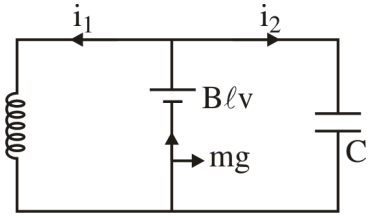
SECTION-I

1. Ans (B,C)

$A = \frac{5}{100}(m) = 5\ cm$

Position of ring at a phase angle of 37° and position of wall at a phase of 53° therefore time period $\pi\sqrt{\frac{M}{K}}$.

3. Ans (B,C)



By applying K.V.L. in the loop

$B\ell v = L\frac{di_1}{dt}$ ...(1)

$i_1 = \frac{B\ell}{L}\int v dt \Rightarrow LI_1 = B\ell x \Rightarrow I_1 = \frac{B\ell x}{L}$

$C(B\ell v) = Q$ (in capacitor)

$i_2 = B\ell Ca$

Force equation on the rod

$mg - B(\ell_1 + \ell_2)\ell = ma$

$mg - Bi_1\ell = ma + Bi_2\ell$

$mg - B\ell\left(\frac{B\ell x}{L}\right) = ma + B^2\ell^2 Ca$

$\frac{1}{m + B^2\ell^2 C}\left[mg - \frac{B^2\ell^2 x}{L}\right] = a$

(represent eq. of SHM)

$\omega^2 = \frac{B^2\ell^2}{2ML} \Rightarrow \omega = \frac{B\ell}{\sqrt{2m\ell}}$

5. **Ans (A,B,D)**

By K.V.L.

$$\frac{Ldi}{dt} - iR + \frac{1}{2}B\omega a = 0$$

$$\frac{Ldi}{dt} + iR = \frac{1}{2}\left(\frac{r_0 Ni}{\ell}\right)\omega a^2$$

$$\frac{di}{dt} = \frac{1}{L}\left[\frac{1}{2}\left(\frac{r_0 Ni}{\ell}\right)\omega a^2 - iR\right]$$

$$\frac{di}{dt} > 0$$

$$\frac{1}{2}\frac{r_0 Ni\omega a^2}{\ell} > ir$$

$$\omega > \frac{2r\ell}{r_0 Na^2}$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (4.08)**

$$I = \frac{25K\omega}{\pi R^2} = \frac{1}{2}\epsilon_0 E^2 C$$

$$E^2 = \frac{50 \times 10^3}{\pi\epsilon_0 \times 36 \times 10^{-4} \times 3 \times 10^8}$$

$$\epsilon^2 = \frac{50 \times 10^3 \times 36 \times 10^9}{36 \times 10^{-4} \times 3 \times 10^8} \Rightarrow \frac{50}{3} \times 10^8$$

$$E^2 = 16 \times 10^8$$

2. **Ans (1.88)**

$$Kx_S + Kx_B = m\omega^2 x_B$$

$$\frac{Kx}{4} + \frac{K21x}{4} = \frac{m\omega^2 21x}{4}$$

$$\frac{22Kx}{4} = \frac{m\omega^2 21x}{4}$$

$$\omega = \sqrt{\frac{22K}{21m}}$$

3. **Ans (2.09 to 2.10)**

$$\int_A^B B \cdot d\ell = \frac{\mu_0 i}{2\pi R} R\frac{\pi}{6}$$

$$\frac{\mu_0 i}{12} \Rightarrow \frac{\mu_0}{6} \Rightarrow \frac{4\pi}{6} \times 10^{-8} \Rightarrow \frac{6.28}{3} = 2.09$$

4. **Ans (2280.00)**

$$\frac{1200 \times 10^3}{4000} = 300\ A$$

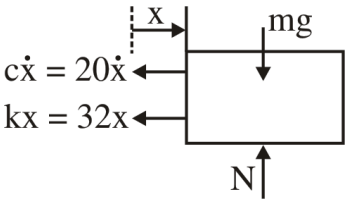
Line loss of power = $(300)^2\ \Omega$

$= (300)^2 \times 15 = 9 \times 10^4 \times 12$ Þ 1080 KW

Power plant power = 1200 KW + 1080 KW

= 2280 KW

5. **Ans (1.50 to 1.56)**



We must first determine whether the system is underdamped, critically damped, or overdamped. For that purpose, we compute the damping ratio ξ.

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{32}{8}} = 2\ rad/s$$

$$\xi = \frac{c}{2m\omega_n} = \frac{20}{2(8)(2)} = 0.625$$

Since $\xi < 1$, the system is underdamped. The damped natural frequency is

$$\omega_d = \omega_n\sqrt{1-\xi^2} = 2\sqrt{1-(0.625)^2} = 1.561\ rad/s$$

6. **Ans (5)**

Applying conservation of flux,

$L_i I_i = L_f I_f$

$\Rightarrow 5 \times 1 = 1 \times I_f$, So $I_f = 5A$

PART-1 : PHYSICS

SECTION-III

1. Ans (1)

$$X_L = \omega L = \frac{5}{\sqrt{3}}(3.6) = 6\sqrt{3}$$

$$X_C = \frac{1}{\omega C} = \frac{1}{\frac{5}{\sqrt{3}}(0.1)} = 2\sqrt{3}$$

$R = 4a$

$$Z = \sqrt{R^2 + (X_L - X_C)^2} = 8 \text{ ohm}$$

e.m.f. leads current by 60°

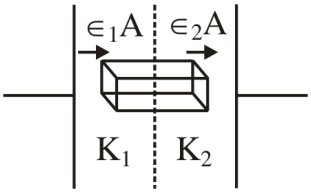
$$I = 2\sin\left(\frac{5}{\sqrt{3}}t - 60\right)$$

$$V_C = 4\sqrt{3}\sin\left(\frac{5}{3}t - 60° - 90°\right)$$

$$C = \frac{\epsilon_0 A}{d} \Rightarrow 0.1 = \frac{\epsilon_0 \times 10^{-4}}{d} \Rightarrow d = \epsilon_0 10^{-3}$$

Electric field in capacitor

$$\frac{V_C}{d} = \frac{4\sqrt{3}}{\epsilon_0 \times 10^{-3}}\sin$$



by applying Gauss Law

$$\left(\frac{K_2 - K_1}{K_1 K}\right)(\epsilon_2 - \epsilon_1)A = \frac{Q_{in}}{\epsilon_0}$$

$$\frac{\sigma}{\epsilon_0} = \left[\frac{2}{4}\right] = \frac{4\sqrt{3}}{\epsilon_0 \times 10^{-3}}\sin\left(\frac{5}{\sqrt{3}} \times \frac{7\pi}{10\sqrt{3}} = 150\right)$$

2. Ans (1)

$$\frac{M}{L} = \frac{Q}{2M} \Rightarrow \vec{M} \text{ (magnitude of cone)} = \left(\frac{\pi R^2}{2}i\right)$$

Torque of magnetic force MB sin θ $\Rightarrow \frac{\pi R^2 iB}{2}$

$$\tau_B > \tau_{mg} \Rightarrow \frac{\pi R^2 iB}{2} = mgR$$

$\Rightarrow i = 2mgBR$

3. Ans (5)

$$X_L = \omega L \Rightarrow 100 \times 80 \times 10^{-3} = 82$$

$$X_C = \frac{1}{\omega C} \Rightarrow \frac{1}{100 \times 2.5 \times 10^{-3}} \Rightarrow 4\Omega$$

$$\frac{1}{R_{eq}} = \frac{1}{8\hat{i}} + \frac{1}{4\hat{i}} \Rightarrow i\left[\frac{1}{8}\right] \Rightarrow R_{eq} = -8\hat{i}$$

$$Z_{eq} = 6\hat{i} - 8\hat{i} \Rightarrow \sqrt{36 + 64} \Rightarrow 10$$

4. Ans (6)

$$\frac{1}{2} \times 8 \times x = \frac{\phi_2 - \phi_1}{2}$$

Area of i – t graph is charge 3C.

$$Q = \frac{\phi_2 - \phi_1}{R} \Rightarrow 6 \text{ weber}$$

5. Ans (2)

When velocity is maximum then net force acting on the conductor is zero.

6. Ans (2)

Initially when the magnetic field is building up, impulses of the force of interaction between the currents induced in the jumpers due to the time varying magnetic field and the magnetic field provide each of the jumpers an inward momentum. Thereafter, each jumper will evetually stop by losing its momentum due to the impulse of force of interaction between the static magnetic field and the current caused by motional emf in the jumper.

$$\frac{Blx_0}{\Delta t} = I \times 2R \Rightarrow I = \frac{Blx_0}{2R\Delta t}$$

Change in momentum $P = IlB\Delta t = B^2l^2x_0 / 2R$

Now at an instant let velocity of conductor is v, so emf is induced in the conductor.

$$I_{in} = \frac{2Blv}{2R}, \quad F = IlB = \frac{B^2l^2v}{R}$$

$$\int F\,dt = \frac{B^2l^2}{R}\int V\,dt = P$$

$$\Rightarrow \int V\,dt = x_0/2$$

PART-2 : CHEMISTRY

SECTION-I

1. Ans (A,B,C,D)

NCERT text book

2. Ans (A,C,D)

(A) Paired base sequence from 5' to 3' will be TGACACTG.

(B) Bases are derived from pyrimidine.

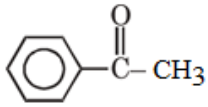
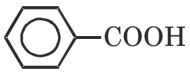
(C) $\left[\begin{matrix}{}^{3'}\text{Sugar}^{5'}\text{– Phosphate}\\ \quad\quad|\\ \quad\text{Base}\end{matrix}\right]_n$

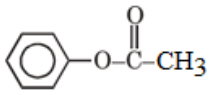
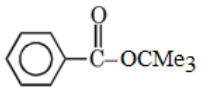
(D) A:::T, G:::::C

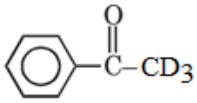
3. **Ans (A)**

NCERT Theory

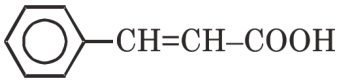
4. **Ans (A,C,D)**

(A)  (B) 

(C)  (D) 

(A) 

(B) Dimerization phenomenon, i < 1.

(C) Perkin reaction 

(148 amu)

(D) One is BAc_2 and another is BAl_1

5. **Ans (A,B)**

(A) Reduction of amides

(B) Carbylamine reaction reduction of isocyanides

(C) Imine formation takes place

(D) Nucleophilic addition not possible

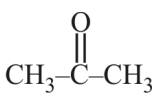
6. **Ans (A,D)**

(A) Hydration depends on extent of partial positive charge & steric hinderance.

(B) Haloform reaction

(i) $CH_3CHO \xrightarrow{CaOCl_2} (HCOO)_2Ca \xrightarrow{\Delta}$ 

(ii) $CH_3COCH_3 \xrightarrow{CaOCl_2} (CH_3COO)_2Ca \xrightarrow{\Delta}$



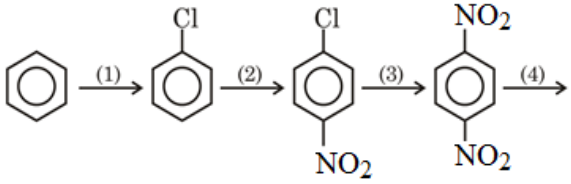
(C) CCl_3CHO gives $CHCl_3$, not iodoform.

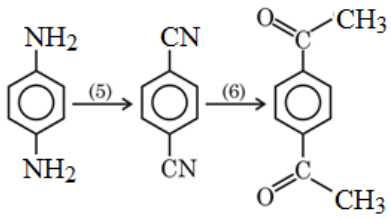
(D) Electronic effects

PART-2 : CHEMISTRY

SECTION-II

1. **Ans (116.64)**





Accounting for yield, moles of product formed

$= 0.90 \times 0.80 \times 1$

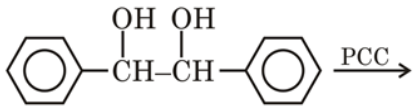
$= 0.72$

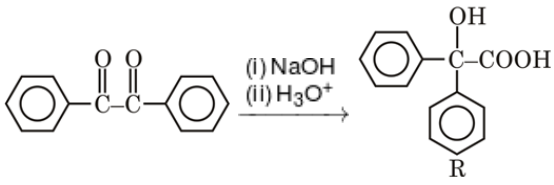
molar mass = 162 g

mass of product obtained = 0.72×162

$= 116.64$ g

2. **Ans (25.33)**





Molar mass = 228 g/mol

DoU = 9

3. **Ans (3.00)**

$$2CH_3COONa + 2H_2O \xrightarrow{2e^-} \underbrace{C_2H_6 + 2CO_2}_{\text{Anode}} + \underbrace{2NaOH + H_2}_{\text{Cathode}}$$

Number of Faradays passed

$$= \frac{9.65 \times 10^{-3} \times 2500}{96500} = 2.5 \times 10^{-4}$$

= number of eq. of C_2H_6 produced.

Equivalent mass of $C_2H_6 = \frac{30}{2} = 15$ g. eq.

$\therefore$ Mass in mg of C_2H_6 (ideally)

$= 2.5 \times 10^{-4} \times 15 \times 10^3$ mg

As current efficiency is 80%,

mass of C_2H_6 produced $= \frac{80}{100} \times 3.75$

$= 3.00$ mg

4. **Ans (10.00)**

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5. **Ans (1.50)**

a = 2 (Aspartic acid, Glutamic acid)

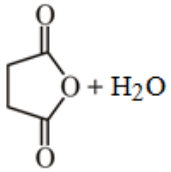
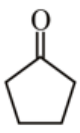
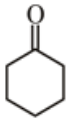
b = 3 (Arginine, Histidine, Lysine)

c = 4 (Phenylalanine, Tyrosine, Tryptophan, Histidine)

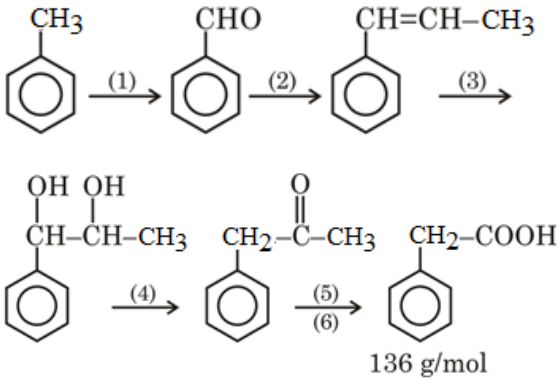
PART-2 : CHEMISTRY

SECTION-III

1. **Ans (5)**

Reactant	Product
$HOOC–(CH_2)–COOH$	$CH_3COOH + CO_2$
$HOOC–(CH_2)_2–COOH$	 $+ H_2O$
$HOOC–(CH_2)_3–COOH$	glutaric anhydride $+ H_2O$
$HOOC–(CH_2)_4–COOH$	 $+ CO_2 + H_2O$
$HOOC–(CH_2)_5–COOH$	 $+ CO_2 + H_2O$

2. **Ans (8)**



136 g/mol

3. **Ans (2)**

Penicillin, ofloxacin-bactericidal

Erythromycin, tetracycline, chloramphenicol-bacteriastatic.

4. **Ans (6)**

Polyethene, Teflon, PVC, PMMA, Polystyrene, Natural rubber,

5. **Ans (4)**

(iv), (v), (vi), (vii)

6. **Ans (3)**

$$R_f = \frac{9}{15} = 0.6$$

PART-3 : MATHEMATICS

SECTION-I

1. **Ans (C)**

The possible powers of 'x' in first 3 brackets are

0, 2, 4, 6 ;

$0, \frac{1}{3}, \frac{2}{3}, 1, \frac{4}{3}, \frac{5}{3}, 2, \frac{7}{3}$;

$0, \frac{1}{3}, \frac{2}{3}, 1, \frac{4}{3}$

To get x^6 we need sum of powers

$\left(4, \frac{2}{3}, \frac{4}{3}\right), (4, 1, 1), \left(4, \frac{4}{3}, \frac{2}{3}\right), \left(4, \frac{5}{3}, \frac{1}{3}\right), (4, 2, 0), (6, 0, 0)$

$${}^3C_2 \cdot {}^7C_2 \, {}^4C_4 + {}^3C_2 \left(-{}^7C_3\right) \cdot {}^4C_3 + {}^3C_2 \, {}^7C_4 \, {}^4C_2 + {}^3C_2 \left(-{}^7C_2\right) {}^4C_1 + {}^3C_2 \, {}^7C_6 \, {}^4C_0 + {}^3C_3 = 43$$

2. **Ans (C)**

$x_1 \; G_1 \; x_2 \; G_2 \; x_3 \; G_3 \; x_4 \; G_4 \; x_5 \; G_5 \; x_6$

$x_1 + \ldots . + x_6 = 15, \; x_1, x_6 \geq 0, \; x_2, \; x_3 \; , x_4, x_5 \geq 2$

$\therefore \; {}^{7+6-1}C_{6-1} = {}^{12}C_5$

$\therefore \; P = \frac{{}^{12}C_5 \cdot 5! \cdot 15!}{20!} = \frac{12! \, 15!}{7! 20!}$

3. **Ans (A,C)**

$$\frac{1}{2^{15}}\sum_{r=0}^{15}(-1)^r\,5^{15-r}\,\frac{30!}{r!\,(30-r)!}\,\frac{(30-r)!}{(15-r)!}\,15!$$

$$=\left(\frac{5}{2}\right)^{15}\sum_{r=0}^{15}\left(-\frac{1}{5}\right)^r\frac{30!}{15!\,15!}\,\frac{15!}{r!\,(15-r)!}$$

$$=\left(\frac{5}{2}\right)^{15}{}^{30}C_{15}\sum_{r=0}^{15}{}^{15}C_r\left(-\frac{1}{5}\right)^r$$

$$=\left(\frac{5}{2}\right)^{15}{}^{30}C_{15}\left(\frac{4}{5}\right)^{15}={}^{30}C_{15}\,2^{15}$$

4. **Ans (B,D)**

Clearly '0' can't be any of the digit

Suppose 1 occurs x_1 times

2 occurs x_2 times

$\vdots$

9 occurs .....x_9 times

$\therefore\ x_1+x_2+\ldots.+x_9=15$

$\therefore$ No of integral sol $={}^{15+9-1}C_{9-1}={}^{23}C_8$

$$=\frac{23!}{8!\,15!}$$

$= 23.\,22.\,3.\,19.\,17$

5. **Ans (B,D)**

$\sqrt{6}-\sqrt{5}>\sqrt{8}-\sqrt{7}>\log_{\sqrt{6}-\sqrt{5}}\sqrt{8}-\sqrt{7}<1$

$\because\ \sqrt{6}-\sqrt{5}\in(0,\ 1)$

Also

$$(\sqrt[3]{2}+\sqrt[3]{5})\left((\sqrt[3]{2})^2+(\sqrt[3]{5})^2-\sqrt[3]{2}.\sqrt[3]{5}\right)$$

$= 2+5=7$

$\therefore\ \log_7^7=1$

$$T_{r+1}={}^{105}C_r\,7^{\frac{105-r}{7}}.\,2^{r/5}={}^{105}C_r\,7^{15}.\,\frac{2^{r/5}}{7^{r/7}}$$

6. **Ans (A,C)**

$$S=\sum_{r=1}^{2023}\sin(rx)\,\cos(2024-r)\,x$$

$$S=\sum_{r=1}^{2023}\sin(2024-r)x.\cos rx$$

$$2S=\sum_{r=1}^{2023}\sin 2024x$$

$$S=\frac{2023}{2}\sin 2024x=\frac{2023}{2}$$

$\sin(2024x)=1$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (0.05)**

$$\sum_{n=4}^{2023}\frac{n+3}{(n+1)!\,(n+3)^2}=\sum_{n=4}^{2023}\frac{1}{(n+1)!\,(n+3)}$$

$$=\sum_{n=4}^{2023}\frac{n+2}{(n+3)!}=\sum_{n=4}^{2023}\frac{(n+3)-1}{(n+3)!}$$

$$=\sum_{n=4}^{2023}\left(\frac{1}{(n+2)!}-\frac{1}{(n+3)!}\right)$$

$$=\frac{1}{6!}-\frac{1}{2026!}$$

2. **Ans (2.00)**

Let $\log_{\sqrt{2}-1}\tan x=t$

$$t\cdot\sqrt{\frac{3}{2}+\frac{1}{t}}=-1\qquad\therefore\ t<0$$

$$\Rightarrow\ \frac{3}{2}+\frac{1}{t}=\frac{1}{t^2}\Rightarrow 3t^2+2t=2$$

$$\Rightarrow 3t^2+2t-2=0\Rightarrow t=\frac{-2\mp\sqrt{28}}{6}$$

$$t=\frac{-1\pm\sqrt{7}}{3}\ \&\ \frac{3}{2}+\frac{1}{t}>0$$

$$\Rightarrow\frac{3}{2}t+1<0\ \Rightarrow\ t<-\frac{2}{3}$$

$$\therefore\ t=\frac{-1-\sqrt{7}}{3}\ ;\ \log_{\sqrt{2}-1}\tan x=\frac{-\left(\sqrt{7}+1\right)}{3}$$

$$\tan x=(\sqrt{2}-1)^{-\left(\frac{\sqrt{7}+1}{3}\right)}=\frac{1}{(0.414)^{\sqrt{7}+1}}$$

3. **Ans (0.00)**

$10-2\sin^2\theta.\cos^2\theta-6\sin\theta\cos\theta\,(\sin\theta+\cos\theta)=0$

$(s^4+c^4)+9\,(s^2+c^2)-6s^2c-6c^2s=0$ (s = sinθ and c = cosθ)

$(s^2-3c)^2+(c^2-3s)^2=0$

$\therefore\ \sin^2\theta=3\cos\theta$ and $\cos^2\theta=3\sin\theta$

$\therefore$ No solution

4. **Ans (6.00)**

$$\frac{1}{\log_{pq}^{q}}-\frac{1}{\log_{pq}^{p}}=\log_q^{pq}-\log_p^{pq}$$

$$=\log_q^p-\log_p^q$$

$$=\sqrt{\left(\log_q^p+\log_p^q\right)^2-4}=45$$

cosec θ = 45 has 6 solutions

5. **Ans (19.00)**

$$\frac{m}{n} = \frac{\frac{7}{11}\cdot\frac{6}{10}\cdot\frac{5}{9}\cdot\frac{4}{8}}{\frac{7}{11}\cdot\frac{4}{10}+\frac{7}{11}\cdot\frac{6}{10}\cdot\frac{5}{9}\cdot\frac{4}{8}+\frac{7}{11}\cdot\frac{6}{10}\cdot\frac{5}{9}\cdot\frac{4}{8}\cdot\frac{3}{7}\cdot\frac{4}{6}+\frac{7}{11}\cdot\frac{6}{10}\cdot\frac{5}{9}\cdot\frac{4}{8}\cdot\frac{3}{7}\cdot\frac{2}{6}\cdot\frac{1}{5}\cdot 1}$$

$$= \frac{7}{26}$$

$\therefore$ n–m = 19

6. **Ans (0.36)**

A = {1, 3, 5, 7, 9}

B = {7}

C = {7, 9, 3, 1}

(A $\cap$ B) $\cup$ C = {7, 9, 3, 1}

$\therefore$ Required probability = 0.11 + 0.14 + 0.03 + 0.08 = 0.36

PART-3 : MATHEMATICS

SECTION-III

1. **Ans (2)**

X ; $n_1p_1 = 30$; $n_1p_1q_1 = 5$

$\therefore q_1 = \frac{1}{6}, p_1 = \frac{5}{6}, n_1 = 36$

Y : $n_2p_2 = 9$; $n_2p_2q_2 = \frac{27}{4}$

$\therefore q_2 = \frac{3}{4}, p_2 = \frac{1}{4}, n_2 = 36$

Now $a^{a+b} = b^{12} \Rightarrow b = a^{\frac{a+b}{12}}$

$\therefore b^{a+b} = a^3 \Rightarrow a^{\frac{(a+b)^2}{12}} = a^3,$

$\therefore\ a+b=6 \text{ and } b^2 = a$

$\therefore b^2 + b = 6 \Rightarrow b = 2, a = 4$

$$\frac{P_1(X=4)}{P_2(Y=2)} = \frac{{}^{36}C_4\left(\frac{5}{6}\right)^4\left(\frac{1}{6}\right)^{32}}{{}^{36}C_2\left(\frac{1}{4}\right)^2\left(\frac{3}{4}\right)^{34}} = \frac{11.17.5^4.2^{35}}{3^{70}}$$

$\therefore$ p = 5, q = 3

2. **Ans (5)**

sec θ + cosec θ = sin θ + cos θ

$\Rightarrow$ (sin θ + cos θ) (1 – sinθ.cosθ) = 0

$\therefore$ tan θ = –1 on sin 2θ = 2

$\therefore \theta = \frac{3\pi}{4}, \frac{7\pi}{4}$

$A = \left\{\frac{3\pi}{4}, \frac{7\pi}{4}\right\}$

$\sqrt{1-\sin\theta} = \cos\theta$, squaring

$\Rightarrow \sin\theta = 0, 1$

$\theta = 0, \pi, 2\pi, 3\pi, \frac{\pi}{2}, \frac{5\pi}{2}$

putting and verifying

$B = \left\{0, 2\pi, \frac{\pi}{2}, \frac{5\pi}{2}\right\}$

$A \cup B = \left\{0, \frac{\pi}{2}, \frac{3\pi}{4}, 2\pi, \frac{7\pi}{4}, \frac{5\pi}{2}\right\}$

$\theta = \frac{3\pi}{4}$ dosen't satisfy C

$\therefore$ n (A $\cup$ B) $\cap$ C) = 5

3. **Ans (3)**

$3 + 3\log_2\cos 6x + 3\log_2\cos x = \log_2(\cos^3 5x + \cos^3 7x)$

$\Rightarrow \log_2(8\cos^3 6x.\cos^3 x) = \log_2(\cos^3 5x + \cos^3 7x)$

$\Rightarrow 8\cos^3 6x.\cos^3 x = \cos^3 5x + \cos^3 7x$

$\Rightarrow (2\cos 6x.\cos x)^3 = \cos^3 5x + \cos^3 7x$

$\Rightarrow (\cos 5x + \cos 7x)^3 = \cos^3 5x + \cos^3 7x$

$(a+b)^3 = a^3 + b^3$

$\Rightarrow 3ab(a+b) = 0$

$\because a + b \neq 0 \quad \therefore a = 0 \text{ or } b = 0$

So cos 5x = 0

$\Rightarrow x = \frac{\pi}{10}, \frac{3\pi}{10}, \frac{\pi}{2}$ but cos 7x, cos 6x, cos x > 0

$\Rightarrow x = 3\pi/10$

or cos 7x = 0

$x = \frac{\pi}{14}, \frac{3\pi}{14}, \frac{5\pi}{14}, \frac{\pi}{2}$ but cos 5x, cos 6x, cos x > 0

$\Rightarrow \frac{\pi}{14}, \frac{5\pi}{14}$

4. Ans (2)

$$\sum x_i = 100 + 70 = 170$$

$$\sigma^2 = \frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n}\right)^2$$

$$\sum x_i^2 = \left(\frac{5}{2}+p\right)^2.20 + \left(\frac{7}{2}\right)^2.20 + \left(\frac{5}{2}-p\right)^2.20$$

$$= 20\left[\frac{99}{4} + 2p^2\right]$$

$$\sum x_i = 20\left[\frac{5}{2}+p+\frac{7}{2}+\frac{5}{2}-p\right]$$

$$= 20\left[5+\frac{7}{2}\right] = 170$$

$$\frac{4}{9} = 20\frac{\left[\frac{99}{4}+2p^2\right]}{60} - \frac{289}{36}$$

$$\frac{305-297}{12} = 2p^2$$

$$\therefore 6p^2 = 2$$

5. Ans (6)

$$f(m) = \sum_{r=1}^{m} \frac{\left(m^r + m^{r-1}\right)\left(1-\frac{1}{r}\right)}{{}^mC_r}$$

$$= \sum_{r=1}^{m} \frac{m^{r-1}(m+1)(r-1)}{r.{}^mC_r}$$

$$= \sum_{r=1}^{m} \frac{m^r}{{}^mC_r} + \frac{m^{r-1}(r-1) - m^r}{r.{}^mC_r}$$

$$= \sum_{r=1}^{m} \frac{m^r}{{}^mC_r} - \frac{m^{r-1}(m-r+1)}{r.\,m!}r!(m-r)!$$

$$= \sum_{r=1}^{m} \frac{m^r}{{}^mC_r} - \frac{m^{r-1}(m-r+1)}{r.{}^mC_r}(m-(r-1))$$

$$= \sum_{r=1}^{m} \frac{m^r}{{}^mC_r} - \frac{m^{r-1}(m-r+1)}{r.\,m!}r!(m-r)!$$

$$= \sum_{r=1}^{m} \frac{m^r}{{}^mC_r} - \frac{m^{r-1}r(r-1)!(m-(r-1))!}{r.\,m!}$$

$$= \sum_{r=1}^{m} \left(\frac{m^r}{{}^mC_r} - \frac{m^{r-1}}{{}^mC_{r-1}}\right)$$

$$= m^m - 1$$

6. Ans (4)

R G B Y P (These pattern should not occur).

The possible combination i.e

B P G Y R

B P R Y G

B R P G Y

B R Y G P

G P B R Y

G Y R B P

G Y R P B

P B R Y G

P G Y R B

R B P G Y

R Y G P B

Y G P B R

Y G P R B

Y R B P G

$$\frac{2}{7}N = \frac{2}{7}.14 = 4$$